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COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #1

Only one sheet of paper 5 x 7 is allowed

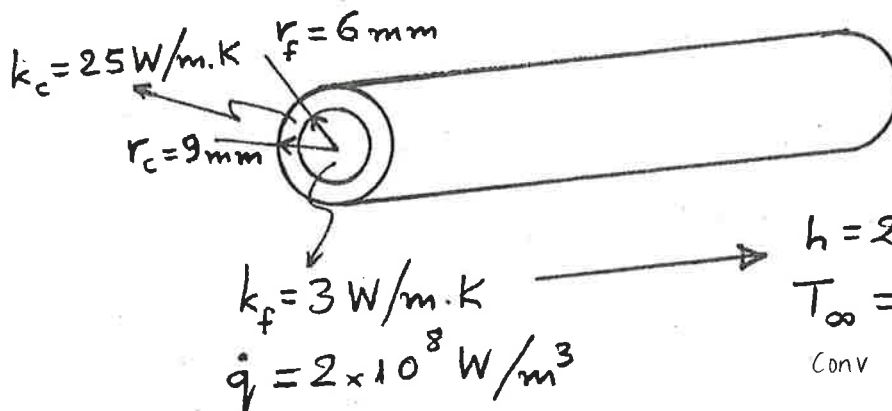
Duration: 50 minutes --- February 10, 2020

Both Problems must be done

Problem 1: (50 points)

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation $\dot{q}$ occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- a) Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
- b) Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?



$h = 2000 \text{ W/m}^2\cdot\text{K}$
 $T_\infty = 300 \text{ K}$
Conv

a) $T_f(r), T_c(r)$ $r_f, k_f, r_c, k_c, \dot{q}, T_\infty, h$

temp distribution cylindrical $T_{s,2} + \Delta T \frac{\ln(r/r_2)}{\ln(r_1/r_2)}$

$$T_f(r) = T_{s,2} + \Delta T \frac{\ln(r/r_2)}{\ln(r_1/r_2)}$$

one-dimensional, steady state...
no heat gen ✓

Heat diffusion eqn

cylindrical $\frac{1}{r} \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) + \frac{1}{r^2} \frac{\partial}{\partial \phi} (k r^2 \frac{\partial T}{\partial \phi}) + \frac{\partial}{\partial z} (k \frac{\partial T}{\partial z}) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$

$$\frac{1}{r} \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) = -\dot{q}$$

$$\int \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) = -\dot{q}r \rightarrow kr \frac{\partial T}{\partial r} = -\frac{\dot{q}r^2}{2} + C_1$$

$$\rightarrow \int \frac{\partial T}{\partial r} = \int \left(-\frac{\dot{q}r}{2k} + \frac{C_1}{kr} \right) \rightarrow T(r) = -\frac{\dot{q}r^2}{4k} + \frac{C_1}{kr} + C_2 \quad \int \frac{C_1}{r} dr = C_1 \ln r$$

B.C.1 $r=0$

B.C.2 $r=f$

$$T(0) = C_2$$

$$T(f) = -\frac{\dot{q}f^2}{4k} + \frac{C_1}{kf} + C_2 \quad \times$$

$$T(f) = -\frac{\dot{q}f^2}{4k} + \frac{C_1}{kf} + T(0) \rightarrow T(f) - T(0) + \frac{\dot{q}f^2}{4k} = \frac{C_1}{kf}$$

$$T_f(r) = -\frac{\dot{q}r^2}{4k} + \left[T(f) - T(0) + \frac{\dot{q}f^2}{4k} \right] + T(0)$$

$$C_1 = \left(T(f) - T(0) + \frac{\dot{q}f^2}{4k} \right) kf^2$$

$$T_c(r) = -\frac{\dot{q}r^2}{4k} + \left[T(f) - T(0) + \frac{\dot{q}f^2}{4k} \right] + T(0) \quad \times$$

b) $q_x'' = \frac{T_{\infty,1} - T_{\infty,2}}{R_{tot}}$

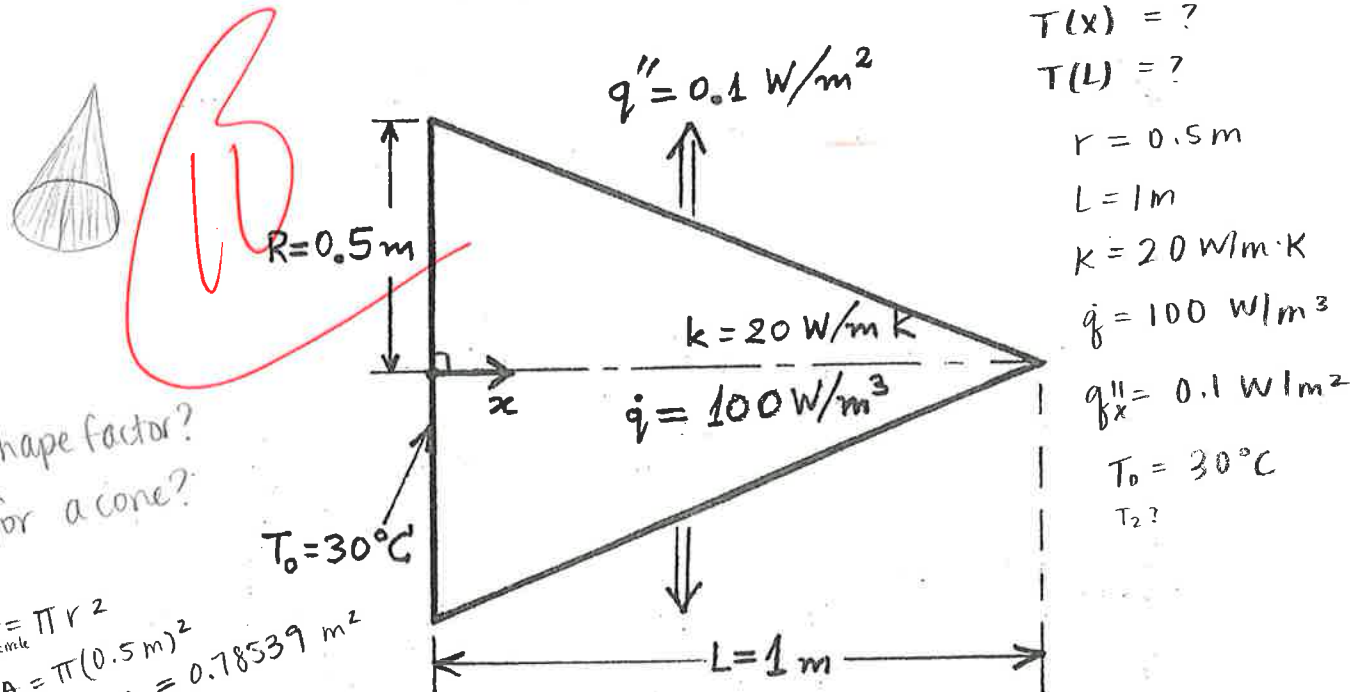
$$q_x'' = h(T_s - T_\infty) \rightarrow (2000 \text{ W/m}^2 \cdot \text{K})(T_s - 300 \text{ K})$$

$$T_f(r) = -\frac{\dot{q}(0.006 \text{ m})^2}{4(3 \text{ W/m} \cdot \text{K})} + \frac{C_1}{(0.006 \text{ m})} + C_2 \quad \times$$

$$T_c(r) = \frac{(-2 \times 10^8 \text{ W/m}^3)(0.009 \text{ m})^2}{4(25 \text{ W/m} \cdot \text{K})} + \frac{C_1}{(0.009 \text{ m})} + C_2$$

Problem 2: (50 points)

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity 20 W/m·K has uniform volumetric heat generation of 100 W/m³. A constant heat flux of 0.1 W/m² is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_0 = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.



shape factor?
for a cone?

$$A = \pi r^2$$

$$A = \pi (0.5\text{ m})^2$$

$$A = 0.78539\text{ m}^2$$

Heat Diffusion Eqn

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

• assume: steady state
- one-dimensional

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{q} = 0 \rightarrow \int \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) = \int -\dot{q} \rightarrow \int k \frac{\partial T}{\partial x} = \int -\dot{q} x + C_1 \rightarrow T(x) = \frac{-\dot{q} x^2}{k 2} + \frac{C_1}{k} x + C_2$$

B.C. 1 $x = 0$

B.C. 2 $x = 1$

$$T(0) = C_2$$

$$T_1 = C_2$$

$$T(1) = \frac{-\dot{q}(1)^2}{k 2} + \frac{C_1(1)}{k} + C_2 \rightarrow \frac{-\dot{q}}{2k} + \frac{C_1}{k} + C_2$$

$$T_2 = \frac{-\dot{q}}{2k} + \frac{C_1}{k} + C_2$$

$$T_2 = \frac{-\dot{q}}{2k} + \frac{C_1}{k} + T_1 \rightarrow 30^\circ\text{C} = \frac{-100\text{ W/m}^3}{2(20\text{ W/m}\cdot\text{K})} + \frac{C_1}{(20\text{ W/m}\cdot\text{K})} + T_1$$

$$q_{x''} = +k \frac{T_1 - T_2}{L} \Rightarrow 0.1\text{ W/m}^2 = 20\text{ W/m}\cdot\text{K} \frac{T_1 - 30^\circ\text{C}}{1\text{ m}} \Rightarrow 0.005\text{ K} = T_1 - 30^\circ\text{C}$$

$$\frac{\text{W}}{\text{m}^2} \frac{\text{m}\cdot\text{K}}{\text{W}} = \text{K}$$

$$T_1 = 30.005^\circ\text{C}$$

$$30^\circ\text{C} = \frac{-100\text{ W/m}^3}{2(20\text{ W/m}\cdot\text{K})} + \frac{C_1}{(20\text{ W/m}\cdot\text{K})} + 30.005^\circ\text{C} \rightarrow -0.005^\circ\text{C} + 2.5 \frac{\text{K}}{\text{m}^2} = \frac{C_1}{20\text{ W/m}\cdot\text{K}} \rightarrow C_1 = 49.9$$

$$C_2 = 30.005$$

$$T(x) = \frac{-q_0 x^2}{2k} + \frac{c_1}{k} x + c_2$$

$$T(x) = \frac{-100 x^2}{2(0.1)} + \frac{49.9}{(0.1)} x + 30.005 \Rightarrow -500x^2 + 499x + 30.005$$

@ radius? so like

B.C.1 $x=0$

$x=0.5$ for $T(x) \uparrow$

$$T(0.5) = \frac{-6(0.5)^2}{k^2} + \frac{c_1}{k}(0.5) + c_2$$

$T(x)?$

$$\text{Dr } T(L) = \frac{-100x^2}{2(0.5)} + \frac{41.95}{(0.5)}x + 30.0025$$

